

SHUNTARO NAKANISHI

Fukuoka, Japan

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EDUCATION

The University of Tokyo

Tokyo, Japan

Admitted to the Master's Program, Department of Mechanical Engineering, for April 2027 entry

- I plan to be enrolled at the University of Tokyo from April through July 2027, then enter a PhD program or a funded master's program abroad if admitted.

Kyushu University

Fukuoka, Japan

Bachelor of Engineering in Aeronautics and Astronautics

Apr 2023 – Mar 2027 (Expected)

- **Relevant Coursework:** Fluid Dynamics, Compressible Fluid Flow, Aerodynamics, Aerospace Propulsion, Numerical Analysis, Differential Equations

RESEARCH INTERESTS

Large-Eddy Simulation, Data-Driven Turbulence Modeling, and Scientific Machine Learning for Computational Fluid Dynamics

RESEARCH EXPERIENCE

Kyushu University, Department of Aeronautics and Astronautics

Undergraduate Researcher (Advisor: Prof. Kenichi Abe)

Apr 2026 – Present

- **Undergraduate Thesis:** *A Study on Nonlinear SGS Models for LES Based on the Tensor Basis Neural Network Using Machine Learning*
- Achieved accuracy comparable to using all 10 of Pope's tensor bases with only the first four ($T^{(1)}-T^{(4)}$) in *a posteriori* simulations of channel flow at $Re_\tau = 395$, reducing velocity overprediction primarily in the buffer and outer layers relative to conventional isotropic models.
- Investigating $T^{(6)}$ as a potential extension of conventional strain-rate-tensor-based SGS models, motivated by its linear dependence on the strain-rate tensor in laminar channel flow.
- Reduced mean-velocity relative L_2 error against DNS from 19.54% with a conventional isotropic model to 3.73% by learning only the coefficient of the isotropic tensor $T^{(6)}$, without incorporating anisotropic tensor terms, using a composite loss function that equally evaluates both the normalized anisotropy tensor and physical stress scale instead of the standard loss function evaluating only the SGS stress anisotropy tensor.

Kyushu University, Faculty of Agriculture

Student Researcher / Lead Developer (Advisor: Prof. Takashi Okayasu)

Sep 2024 – Mar 2025

- **Research Project:** *Generative AI Assistant for Beginner Farmers* (Funded by the Fukuoka Mitou Program, a competitive government-backed innovation program for young software developers)
- Developed a GPT-4-based agricultural assistant that progressively personalizes strawberry cultivation recommendations to individual farms as observations accumulate, addressing the limitations of one-size-fits-all guidance by integrating IoT sensor telemetry, agronomic knowledge, and growers' multiple-choice responses.
- Earned the "Yoka Creator" certification by achieving the top evaluation among 14 funded projects, independently driving the project end-to-end from initial concept to system implementation and evaluation, and securing an industry partnership with SERAKU Co., Ltd. for commercial integration.

WORK EXPERIENCE

SERAKU Co., Ltd.

Tokyo, Japan

Software Engineering Intern

Apr 2025 – Mar 2026

- Enabled the production deployment of the Fukuoka Mitou AI assistant into SERAKU's commercial IoT platform, *Midori Cloud*, by designing backend APIs and data pipelines that connect real-time agricultural sensor telemetry with LLM reasoning.

HONORS, AWARDS, AND GRANTS

Fukuoka Mitou Project Grant

Sep 2024 – Feb 2025

Secured JPY 500,000 in competitive project funding from the Fukuoka Mitou Program for “Development of an AI-based Agriculture Assistant for Beginner Farmers.”

“Yoka Creator” Certification

Feb 2025

Earned certification as a top-tier creator by the Fukuoka Mitou Program, receiving the highest evaluation among 14 funded software projects.

Noshiro Space Event Awards

Oct 2023, Nov 2024

As a member of PLANET-Q, received the Autodesk Award and Mitsubishi Heavy Industries (MHI) Award (2023, 2024) and the Photron Award (2024) for hybrid rocket development at Japan’s largest student rocket competition.

CONFERENCE PRESENTATIONS

Nakanishi, S., Kihara, H., and Abe, K., “A Study on Nonlinear SGS Models for LES Based on the Tensor Basis Neural Network Using Machine Learning,” *The Japan Society for Aeronautical and Space Sciences (JSASS) Western Branch Conference*, Fukuoka, Japan, Nov 2026 (Scheduled for oral presentation).

Nakanishi, S., Sato, S., Okayasu, T., and Mochida, K., “Development of an AI to Support Beginner Farmers by Integrating Environmental Data and Farmers’ Insights,” *Annual Conference of the Japanese Society of Agricultural Informatics*, Kyoto, Japan, May 2025 (Oral presentation).

TECHNICAL SKILLS

Programming & Tools: Python, C/C++, Git, Linux, OpenFOAM

Fluid Dynamics & Computation: Computational Fluid Dynamics (CFD), Large-Eddy Simulation (LES), Turbulence Modeling

Hardware & Embedded Systems: PCB Design, Microcontrollers, CAN Bus, LoRa Telemetry, Serial Communication

EXTRACURRICULAR ACTIVITIES

PLANET-Q (Kyushu University Space Development Club)

Apr 2023 – Present

Avionics Team Member

- Designed the club’s first distributed CAN bus network to improve avionics noise immunity and communication reliability, contributing to successful separation mechanism actuation, full vehicle recovery, and complete flight data logging across launches reaching apogees up to 3,700 m.
- Contributed to consistent combustion characteristics and fail-safe operation across two static firing tests (video) by engineering embedded control electronics and valve actuation systems for precise oxidizer flow regulation and emergency depressurization.

CERTIFICATIONS

Applied Information Technology Engineer

Jan 2025

Ministry of Economy, Trade and Industry / IPA

- Advanced-level national IT certification in Japan, certifying applied technical competence across computer systems architecture, networking, databases, information security, and software design.